

Do topological and quantum-inspired molecular representations add value? Evidence from African natural-product-inspired antimalarial chemical space

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Abstract

Topological and quantum-inspired molecular representations are increasingly proposed for drug discovery, but their predictive value relative to classical fingerprints on natural-product chemical space remains unclear. We benchmark a persistent-homology Topological Fingerprint (TFP), a Tucker-compressed Tensor Network Embedding (TNE), and simulated Quantum Kernel Scores (QKS) against classical baselines on 19 849 computationally generated candidates inspired by African natural-product antimalarials, with computationally predicted activity labels. Persistent homology decomposes the scaffold paradox into conserved H_1 ring features and divergent H_0 connectivity: 92.6% of expanded molecules are whole-molecule Tanimoto-distinct from seeds while 69.3% retain seed scaffolds. TNE compresses molecular tensors to 192 elements at $6.1\times$ real-atom compression (reconstruction error 0.113) and retains docking-score-predictive information, competitively on PfDHFR ($R^2 = \mathbf{0.473}$ versus 0.461 for ECFP4) though below ECFP4 on PfATP4 and PfCRT. For activity prediction

($n = 19\,836$, 5-fold cross-validation), ECFP4 achieves AUC 0.948 under random splitting and 0.822 when folds are grouped by Bemis–Murcko scaffold, the panel spanning only 632 scaffolds; the hybrid (TFP+TNE+QK) reaches 0.888, with QKS the principal contributor within that hybrid ($\Delta = -0.040$), which nonetheless remains below ECFP4. Stacking TFP and TNE onto ECFP4 changed AUC by $-0.000\,2$ ($p = 0.084$). The quantum kernel was not detectably different from a tuned RBF internally ($p = 0.060$) and performed lower on the ChEMBL label-source-shift panel (AUC 0.817 versus 0.847; $p = 0.021$). These representations add diagnostic and interpretive information under the tested protocols, but not predictive performance beyond ECFP4.

Contribution: We introduce three quantum-inspired molecular descriptors: persistent-homology TFP, Tucker-compressed TNE, and simulated QKS; and evaluate them on a computationally generated panel inspired by African natural-product antimalarials using a shared 5-fold cross-validation protocol. The analysis (i) decomposes the scaffold paradox of generative expansion into conserved H_1 ring topology and divergent H_0 connectivity, (ii) quantifies information retention after $6.1\times$ tensor compression, and (iii) compares a quantum kernel with a gamma-tuned RBF baseline within the primary panel ($p = 0.060$) and after transfer to a separately labelled ChEMBL panel, where QKS performance was lower than RBF. Under these conditions, the quantum-inspired representations did not improve predictive performance over the strongest classical baseline.

Highlights:

- Persistent homology decomposes the scaffold paradox into conserved H_1 ring topology and divergent H_0 connectivity
- Tensor Network Embedding achieves $6.1\times$ real-atom compression; retains docking-score-predictive information but degrades on external data
- Quantum kernel did not differ detectably from tuned RBF internally; ChEMBL transfer reveals a performance gap
- Hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888; ablation shows QKS contributes most in this configuration, whereas adding TNE does not improve it
- All AUCs above use random cross-validation; grouping folds by Bemis–Murcko scaffold costs every descriptor 0.086–0.196 AUC, ECFP4 falling from 0.948 to 0.822
- Stacking TFP and TNE onto ECFP4 gives no detectable gain ($\Delta\text{AUC} = -0.000\,2$, $p = 0.084$); every feature-level fusion performs worse than ECFP4 alone
- All activity labels are computational predictions from Ersilia ML models, not experimental measurements

Keywords: persistent homology, tensor networks, quantum kernel methods, molecular fingerprints, scaffold paradox, antimalarial drug discovery, natural-product-inspired chemical space

1 Introduction

Malaria remains a leading cause of mortality in sub-Saharan Africa, with 247 million cases and 619 000 deaths in 2023 [1]. The spread of artemisinin-resistant *Plasmodium*

falciparum from Southeast Asia to Africa [2, 3] creates urgent demand for new chemical scaffolds. African natural products are a historically validated but computationally underexplored source of such scaffolds: quinine and artemisinin derive from natural products, yet African natural products remain underrepresented in public databases [4, 5]. Computational screening of natural-product-derived chemical space depends critically on the molecular representations available to machine learning models.

Extended-Connectivity Fingerprints (ECFP) encode local atom environments into fixed-length bit vectors and have served as the standard descriptor for two decades [6]. These representations perform well for synthetic drug-like molecules whose scaffolds are well represented in training data. Natural product-derived compounds, however, occupy distinct regions of chemical space: higher sp^3 fractions, more chiral centres, more complex ring systems, and greater three-dimensional character [7]. Classical fingerprints may not capture the topological features (bridged rings, molecular cavities, higher-order atomic interactions) that distinguish these molecules.

Three frameworks from physics and topology address orthogonal limitations of classical fingerprints. Topological Data Analysis uses persistent homology to identify shape features persisting across spatial scales, producing persistence diagrams that encode connected components (H_0), holes (H_1), and voids (H_2) [8–10]. Tensor networks compress high-dimensional data by exploiting low-rank structure [11, 12], providing physically interpretable compression that retains scaffold-relevant information. Quantum kernel methods map data into high-dimensional feature spaces through parameterised circuits [13, 14], probing non-linear feature interactions inaccessible to linear or radial-basis comparisons. Each framework captures a different aspect of molecular structure: TDA captures global topology invisible to local substructure hashes; tensor networks provide information-preserving compression; quantum kernels probe Hilbert-space geometry.

We therefore ask four linked questions: (i) can persistent homology resolve the apparent conflict between whole-molecule novelty and scaffold conservation; (ii) does tensor-network compression retain information relevant to computational docking scores; (iii) does a quantum-kernel model add value relative to a tuned RBF comparator; and (iv) does combining these representations improve activity prediction over ECFP4? We address these questions on a common library and distinguish predictive performance from structural diagnosis and information retention. This distinction is important because computational MoA analyses span different biological levels, from compound–target bioactivity to pathway and cellular phenotypes, which are complementary rather than interchangeable evidence types [15]. We compare against ECFP4, MACCS, and other classical baselines under rigorous cross-validation, and assess transfer under a separately sourced ChEMBL label panel without claiming molecule-level independence.

2 Methods

2.1 Dataset

The library consists of 65 856 molecules generated from 396 African natural-product and 454 synthetic-drug seeds via CHEESE similarity search [16] and STONED-SELFIES expansion [17]. The representation analyses use a primary 19 849-molecule panel, of which 19 836 molecules have complete TNE embeddings. Binary activity labels were obtained from the Ersilia eos80ch model (threshold 0.5); the primary panel is 74.2 % active and 25.8 % inactive. For a label-source-shift analysis, 22 447 antimalarial compounds were assembled from ChEMBL 34 ($\text{IC}_{50} \leq 10 \mu\text{M}$ = active, $> 10 \mu\text{M}$ = inactive). ChEMBL is independent of the primary eos80ch label source, but exact molecule-level disjointness from the internal panel was not established; this analysis is therefore a transferability assessment, not an independent molecule-disjoint hold-out validation.

2.2 Classical fingerprint baselines

ECFP4 fingerprints were generated as Morgan fingerprints (radius 2, 2048 bits) using RDKit [18]. MACCS 167-bit structural keys, atom-pair (AP), binding-pattern (BPF), functional-class (FCFP4), and 2D-pharmacophore (PHCO) fingerprints were computed using standard RDKit implementations. All classical fingerprints serve as baselines for the quantum-inspired methods.

2.3 Topological Fingerprint

2.3.1 Molecular graph construction

Each molecule was converted to a 3D conformer using RDKit’s ETKDG generator (random seed 42). A weighted undirected graph $G = (V, E, w)$ was constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij} \left(1 + \frac{Z_i Z_j}{100} \right), \quad (1)$$

where d_{ij} is the Euclidean distance and Z_i, Z_j are atomic numbers. The atomic number weighting emphasises interactions involving heavier atoms, which contribute more strongly to molecular recognition.

2.3.2 Persistent homology

Vietoris–Rips persistence diagrams were computed up to homology dimension 2 using Ripser.py [19]. The topological representation contains persistence summaries for each homology dimension $d \in \{0, 1, 2\}$, including persistence entropy

$$H_d^{\text{ent}} = - \sum_i p_i \log p_i, \quad p_i = \frac{\text{pers}_i}{\sum_j \text{pers}_j}, \quad (2)$$

the count of features with persistence exceeding $0.5 \text{ \AA}$, the maximum persistence, and the mean persistence. The primary activity analysis uses a 78-feature representation obtained by combining these homology summaries with persistence-image and Betti-curve components; the 12-feature core is evaluated separately as TFP-12 in a descriptive topological comparison. Unlike ChemGraphX [20], which computes graph-theoretic indices, our TFP captures geometric persistence from 3D conformations.

2.4 Tensor Network Embedding

2.4.1 Molecular tensor construction

For each molecule, a 3D tensor $T \in \mathbb{R}^{N \times F \times 3}$ was constructed with $N = 100$ maximum atoms (zero-padded), $F = 10$ per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity, hybridization, mass, electronegativity, bond count, ring membership), and 3 Cartesian coordinates: $T_{i,f,c} = x_{i,f} r_{i,c}$.

2.4.2 Tucker compression

Each tensor was compressed using Tucker decomposition [21] with mode ranks $(d, d, 3)$ where $d = 8$ is the bond dimension:

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \quad (3)$$

where $\mathcal{G} \in \mathbb{R}^{d \times d \times 3}$ is the core tensor. The descriptor is the flattened core ($d^2 \times 3 = 192$ elements). For a molecule with N actual atoms, the real compression ratio is $\text{CR}_{\text{real}}(N) = 10N/d^2$; at $d = 8$ and mean $N \approx 39$, this gives $\text{CR}_{\text{real}} \approx 6.1 \times$. Reconstruction error was quantified as the relative Frobenius norm (mean 0.113 across 19836 valid molecules).

2.5 Quantum Kernel Score

Quantum kernels were implemented using PennyLane [22] with a 6-qubit statevector simulator. The kernel value $K(x_1, x_2) = |\langle \phi(x_1) | \phi(x_2) \rangle|^2$ was computed from PennyLane’s IQPEmbedding template. Input vectors were Morgan fingerprint atom-pair counts (radius 2, 2048 features) reduced to 6 dimensions by UMAP (Jaccard metric, random_state=42), then scaled to $[-1, 1]$. The QKS is the AUC achieved by an SVM using the quantum kernel matrix.

2.6 Hybrid descriptor

The hybrid feature vector combines TFP, TNE, and QKS by concatenation:

$$h_i = [\tilde{t}_i, \tilde{e}_i, q_i], \quad (4)$$

where $\tilde{t}_i$ is min-max normalised TFP, $\tilde{e}_i$ is normalised TNE, and $q_i \in \mathbb{R}^{30}$ is the quantum kernel PCA component vector ($n_{\text{kPCA}} = 30$). Random Forest is invariant

to per-component monotone scaling, so no explicit fusion weights are needed. Hyperparameters were compared over 60 combinations using a 200-molecule development subset, and the selected configuration was evaluated on the complete analysis panel (bond dimension 6, 1 repeat, 30 kernel-PCA components). Because the development subset was not held out from this evaluation, the resulting hybrid AUC may be optimistic and should be interpreted as an upper bound for this configuration.

2.7 Evaluation protocol

Descriptor representations were evaluated with Random Forest classifiers (200 trees) under 5-fold stratified cross-validation; the kernel methods (QKS and its RBF and linear baselines) were evaluated with SVMs, which accept a precomputed kernel directly. Descriptor and kernel AUCs are therefore not a like-for-like model comparison. SVM results for the topological vectorisations are reported separately in the Supplementary Material. For the quantum kernel benchmark, the RBF gamma was optimised via inner cross-validation on $\{0.5, 1.0, 2.0, 5.0\}$. Multiplicity was handled by declaring a single confirmatory family and correcting it once. That family comprises the 12 tests supporting the comparative claims on the primary panel: the seven classical descriptors against ECFP4, the hybrid against ECFP4, the quantum kernel against a linear kernel, and the three ECFP4-inclusive fusion arms. Holm–Bonferroni adjustment was applied across those 12 at $\alpha = 0.05$, and adjusted values are reported in the Supplementary Material. The ChEMBL label-source-shift transfer and the secondary correlation analyses lie outside this family; they are reported as exploratory, with uncorrected p -values and effect sizes. For the ChEMBL transfer analysis, we additionally report a corrected-resampled paired t approximation (Nadeau–Bengio variance correction, $df = 4$) and BH-FDR; the naive fold-level t test is shown only for comparison because the training sets overlap. Performance was measured by AUC–ROC, accuracy, and macro F1. Bootstrap 95 % confidence intervals for principal AUC differences are provided in the Supplementary Material.

2.8 Software

Ripser.py [19], TensorLy 0.9.0 [21], PennyLane 0.45.1 [22], RDKit 2025.03.6 [18], UMAP 0.5.12, scikit-learn 1.9.0, NumPy 2.4.6, SciPy 1.17.1. All stochastic procedures — cross-validation splits, Random Forest fitting, UMAP reduction, and bootstrap resampling — were fixed with the seed 42.

3 Results

3.1 Scaffold decomposition by persistent homology

Structural expansion of 396 African natural-product and 454 synthetic-drug seeds produced 65 856 candidates that are 92.6 % Tanimoto-distinct from the seeds yet retain 69.3 % of their Bemis–Murcko scaffolds; the library and these expansion statistics are those of the companion chemical-space study [23]. This apparent contradiction, high whole-molecule diversity with high scaffold conservation, arises because STONED-SELFIES mutations primarily permute side-chains while preserving cyclic cores. TDA

provides a principled decomposition of this phenomenon (Fig. 1). H_1 features (ring topology) are conserved between seed and expanded sets (mean H_1 count 3.61), reflecting the STONED-SELFIES expansion’s preservation of cyclic macro-structures. H_0 features (connectivity) diverge substantially, reflecting the expansion’s permutation of side-chains and functional groups. This H_1/H_0 decomposition is independently confirmed by scaffold-only Tanimoto (0.379) being $1.84\times$ higher than whole-molecule Tanimoto (0.206).

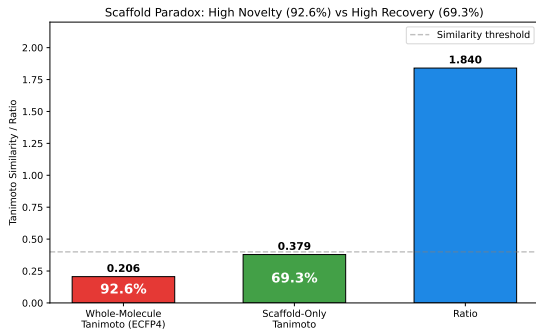


Fig. 1 Scaffold paradox resolution via persistent homology. (A) H_1 persistence distributions for seed natural products versus structurally expanded candidates. (B) H_0 persistence distributions. H_1 preservation with H_0 divergence explains the 92.6 % Tanimoto distinction versus 69.3 % scaffold recovery.

3.2 Binding-relevant information in tensor network compression

Tucker decomposition at bond dimension $d = 8$ yielded $6.1\times$ real-atom compression (mean 39 atoms; 192-element core) with mean relative reconstruction error 0.113. TNE clustering produced a higher Silhouette coefficient (0.35) than ECFP4 (0.18) and the VAE latent representation (0.229) under the specified $k = 484$ KMeans analysis. In cross-validated regression against Tartarus docking scores from the same molecular set, TNE achieved $R^2 = 0.473$ for PfDHFR, compared with $R^2 = 0.461$ for ECFP4; ECFP4 was stronger for PfATP4 and PfCRT. These results support information retention in the analysed panel, but do not establish prospective generalisation or biological binding prediction. The docking structures (1SYH, 6Y2F, 4LDE, 2F6I) serve as an external chemoinformatic validation oracle and are distinct from the refined models (7F3Y, 6UKJ, 9N10) used for molecular dynamics in companion studies.

3.3 Activity prediction benchmark

Table 1 presents activity-prediction performance on the 19836-molecule complete-embedding set. ECFP4 achieved the highest AUC (0.948), followed by AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905), PHCO (0.896), the 78-feature TFP (0.876), and TNE (0.722). The ranking indicates that local atom environments are

particularly informative for these model-derived binary labels, whereas topological and compressed representations provide complementary rather than dominant signal. A descriptive comparison of TFP-12, persistence statistics, persistence images, and Betti curves is provided in the Supplementary Material. These representations were evaluated with different label definitions, classifiers, or sample sizes and are not directly comparable with the primary benchmark.

Table 1 Activity-prediction performance and hybrid ablation. ECFP4, classical fingerprints, TFP, TNE, and the hybrid use the complete-TNE intersection ($n = 19\,836$, 5-fold stratified Random Forest cross-validation). The QKS row is a result from the full representation panel ($n = 19\,849$, 5-fold SVM cross-validation with a precomputed 6-qubit kernel); it is not a like-for-like Random Forest comparison. All values are means across folds. The QKS row is not assigned a Δ AUC versus ECFP4 because it uses a different panel and classifier.

Method	AUC	Accuracy	F1	Δ AUC vs. ECFP4
ECFP4 (baseline)	0.948	0.896	0.931	—
AP	0.940	0.888	0.927	−0.008
BPF	0.939	0.889	0.928	−0.009
FCFP4	0.918	0.869	0.914	−0.029
MACCS	0.905	0.863	0.909	−0.043
PHCO	0.896	0.854	0.904	−0.052
TFP (TDA)	0.876	0.838	0.897	−0.072
TNE ($d = 8$)	0.722	0.756	0.857	−0.226
QKS (6-qubit; SVM, $n = 19\,849$)	0.823	0.819	0.886	—
Hybrid (TFP+TNE+QK)	0.888	0.843	0.900	−0.060
<i>Ablation study (Hybrid minus one component):</i>				
Hybrid − TFP	0.874	0.837	0.895	−0.014
Hybrid − TNE	0.899	0.850	0.903	0.011
Hybrid − QKS	0.847	0.820	0.887	−0.040

3.4 Hybrid descriptor and quantum kernel comparison

The internal hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888 ± 0.007 , significantly below ECFP4 ($p < 0.0001$) but above every standalone quantum-inspired descriptor. Ablation reveals an asymmetric contribution structure: QKS is the principal positive contributor ($\Delta = -0.040$), TFP contributes marginally ($\Delta = -0.014$), while removing TNE slightly improves AUC ($\Delta = +0.011$). The negative TNE contribution is consistent with its lower standalone AUC (0.722) and indicates that adding TNE did not improve this hybrid configuration. The hybrid’s measurable contribution was therefore concentrated in the QKS kernel-PCA components; the hybrid is not recommended over ECFP4 alone for primary screening on this panel.

The quantum-kernel benchmark at two panel sizes (Supplementary Material) did not detect a difference between the quantum kernel and a gamma-tuned RBF baseline ($n = 5\,000$: $p = 0.419$; $n = 19\,849$: $p = 0.060$), while the quantum kernel significantly outperformed the linear kernel ($p \leq 0.0006$). Under the ChEMBL label-source shift (22 447 compounds; Supplementary Material), QKS performance was lower than

RBF (AUC 0.817 versus 0.847; corrected-resampled $p = 0.021$, BH $q = 0.021$). The comparison is limited by overlapping training sets and does not establish quantum advantage.

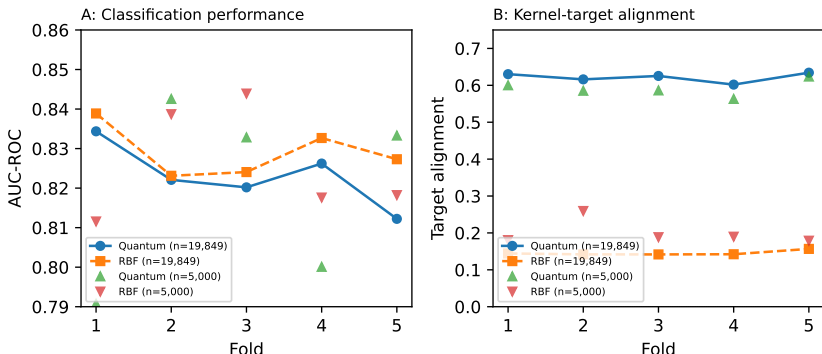


Fig. 2 Quantum kernel benchmark across two panel sizes. (A) AUC-ROC per fold for quantum kernel and gamma-tuned RBF across 5-fold stratified cross-validation at $n = 19\,849$ and $n = 5\,000$. Quantum and RBF overlap at all folds (quantum = 0.823 ± 0.008 , RBF = 0.829 ± 0.007 ; paired $t = -2.59$, $p = 0.060$, ns). (B) Kernel-target alignment per fold. The quantum kernel achieves substantially higher alignment (0.622 ± 0.013) than RBF (0.145 ± 0.006), yet this geometric advantage does not translate into classification performance ($\Delta\text{AUC} = -0.006$).

3.5 Fusion with a classical fingerprint

The hybrid of Table 1 combines the three quantum-inspired descriptors with one another and excludes ECFP4, so it does not test whether they carry information the classical fingerprint lacks. We therefore evaluated ECFP4-inclusive fusion directly, under the protocol of Table 1. A stacked ensemble: a Random Forest per representation, combined by logistic regression over out-of-fold probabilities generated by an inner cross-validation, and therefore the variant least exposed to feature dilution; reached AUC 0.9473 ± 0.0043 , against 0.9475 ± 0.0045 for ECFP4 alone (paired $t = -2.29$, $p = 0.084$). Fusions that mixed the feature blocks directly were worse: concatenation 0.9073 ($\Delta = -0.040$, $p < 0.0001$) and per-block feature selection 0.8958 ($\Delta = -0.052$). The deficit grew as the 2048-bit fingerprint was displaced further, which indicates dilution of a sparse fingerprint rather than interference between representations. For this endpoint, TFP and TNE add no predictive information to ECFP4.

3.6 Transfer under a ChEMBL label-source shift

All descriptor methods were evaluated on the ChEMBL label-source-shift panel (22 447 compounds, 5-fold RF; Supplementary Material). ECFP4 achieved AUC 0.960 ± 0.004 , slightly exceeding its internal value (0.948). TFP achieved AUC 0.864, TNE 0.645, and the TFP+TNE descriptor hybrid 0.800. The primary analysis included the 351 TNE embedding failures (1.6%) as zero-vector penalties; results were also

calculated for the 22 096 complete cases. The complete-case analysis ($n = 22\,096$) gave ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798; the ranking was unchanged. These results assess transfer across label sources, not molecule-disjoint generalisation. Because training sets overlap across folds, the corrected-resampled results are descriptive; descriptor RF and kernel SVM estimates are not direct cross-classifier comparisons.

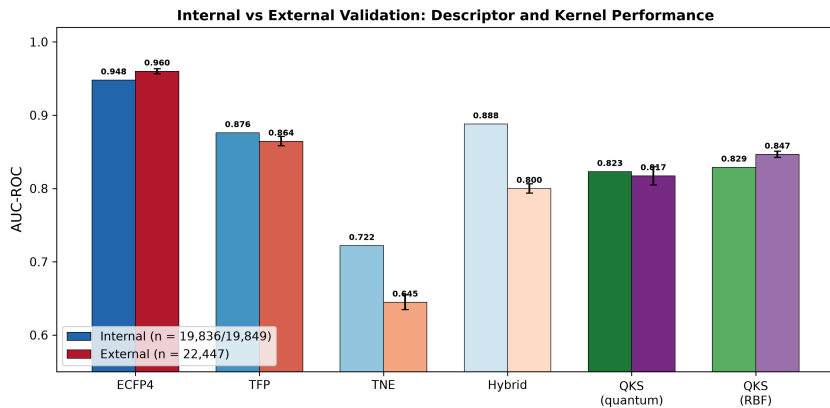


Fig. 3 Internal versus ChEMBL label-source-shift transfer performance. Descriptor methods were evaluated with Random Forest; kernel methods with SVM. Error bars: standard deviation across 5 folds. The ChEMBL panel ($n = 22\,447$) is independent of the eos80ch label source, but molecule-level disjointness from the primary panel was not established; the two classifier families are therefore shown descriptively rather than as a direct comparison.

3.7 Topological features and multi-target docking profiles

Across $N = 17\,011$ molecules docked against four *Plasmodium falciparum* targets (PfDHFR/1SYH, PfCRT/4LDE, PfATP4/6Y2F, PfClpP/2F6I) using AutoDock Vina with target-specific grid configurations (see Supplementary Material for docking parameters and validation), H_0 count ($\rho = -0.248$, $p < 10^{-137}$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$) negatively correlate with the number of targets bound (≥ 2 targets at $\Delta G \leq -7.0 \text{ kcal mol}^{-1}$). Lower values of these features were associated with a larger number of targets meeting the docking threshold. Because the endpoint is a docking-derived count and the associations are weak in magnitude, this result is hypothesis-generating rather than evidence of a mechanistic polypharmacology rule. The association between H_1 count and resistance-resilience scores (RRS), defined in the companion resistance-resilience study [24], is examined in the Supplementary Material; after controlling for molecular size and structural covariates, the association is size-mediated and does not support an independent topological predictor.

4 Discussion

The results answer the four questions posed in the Introduction: persistent homology resolves the scaffold pattern descriptively, TNE retains some docking-related information on its derivation panel, QKS is competitive with but not superior to tuned RBF, and the hybrid does not surpass ECFP4. The scientific value of these representations is therefore diagnostic and interpretive under the tested conditions, not a claim of universal predictive improvement.

4.1 Persistent homology as a diagnostic for generative-model evaluation

The apparent contradiction between 92.6 % Tanimoto distinction and 69.3 % scaffold recovery arises because STONED-SELFIES mutations permute side-chains and functional groups (diverging H_0 connectivity) while preserving cyclic macro-structures (conserving H_1 ring topology). This decomposition has practical implications for generative-model evaluation: scaffold-only Tanimoto (0.379) is $1.84\times$ higher than whole-molecule Tanimoto (0.206), consistent with H_1 persistence capturing ring-level structure that is not represented by whole-molecule similarity alone. TFP therefore provides an interpretable diagnostic for assessing whether molecular-generation protocols preserve ring systems; it complements rather than replaces standard fingerprint-based diversity metrics. We note that this decomposition is consistent with the known tendency of STONED-SELFIES to preserve cyclic cores while permuting side-chains; TDA provides a quantitative, topology-aware framework for measuring this property.

4.2 Tensor-network compression: retention and transfer fragility

Tucker decomposition exploits the fact that molecular tensors are approximately low-rank: the $6.1\times$ real-atom compression retains information sufficient to predict docking scores on the same molecular set (PfDHFR $R^2 = 0.473$, competitive with ECFP4’s 0.461) while providing a fixed-length, physically interpretable embedding. The compression is not uniform across chemical space: molecules with common ring systems compress better (reconstruction error 0.098) than structurally atypical molecules (0.137), and the atom-coordinate factor matrix correlates with molecular volume ($\rho = 0.71$). The higher TNE Silhouette coefficient (0.35) than those of ECFP4 (0.18) and the VAE latent representation (0.229) indicates greater cluster separation under this analysis. Whether that separation improves virtual-screening enrichment was not tested here. We emphasise that the Tartarus docking information-retention analysis uses the same molecular set used to derive the embeddings, not compounds from an independent chemical distribution; a molecule-disjoint prospective transfer analysis would require an audited non-overlapping panel and, ideally, an independently generated activity endpoint.

4.3 Kernel calibration, not quantum-ness, governs performance

With the 6-qubit circuit and a tuned RBF baseline, the quantum kernel did not show a performance advantage. Internally, the difference was not detected ($p = 0.060$ at $n = 19\,849$); under the ChEMBL label-source shift, QKS was lower (AUC 0.817 versus 0.847; corrected-resampled $p = 0.021$, BH $q = 0.021$). Overlapping training sets limit inference from the cross-validation comparison. This is consistent with recent matched-baseline studies: QBioFusion-QSAR [25] found quantum multiple-kernel learning improved small-data classification only for individual molecules, with no mean MCC gain, and parameter-matched comparisons [26] attribute topology-aligned architecture gains to inductive bias rather than quantum-ness.

4.4 Distribution shift exposes tensor network fragility

TNE showed the largest AUC decrease, from 0.722 in the primary analysis to 0.645 on the ChEMBL label-source-shift panel and is consistent with sensitivity to the changed data-generating process. The 351 embedding failures (1.6%) were retained as zero-vector failure penalties in the ITT analysis. The complete-case sensitivity analysis ($n = 22\,096$) gave nearly identical values: ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798, compared with ITT values of 0.960, 0.864, 0.645, and 0.800, respectively. Thus, the qualitative ranking is not driven by the failed embeddings. The corrected-resampled tests use one five-fold partition with overlapping training sets and are interpreted descriptively. Tensor-network descriptors should therefore be evaluated on audited transfer panels before deployment, and embedding failures should be reported transparently. TFP transferred less poorly ($\Delta = -0.011$) than TNE, indicating greater apparent robustness to this distribution shift.

4.5 Limitations

Several limitations constrain interpretation. First, TFP does not outperform ECFP4 in activity prediction; its contribution is primarily diagnostic and may be concentrated in the high-complexity tail (molecules with ≥ 3 fused rings or ≥ 2 stereocentres). Second, activity labels are computational predictions from the Ersilia eos80ch model, so AUC values and claims about activity or polypharmacology quantify prediction of model outputs rather than experimental biological activity. Third, the ChEMBL analysis pools heterogeneous *P. falciparum* assays using a single IC_{50} threshold without assay-specific benchmarking or identification of individual assay IDs, targets, or protocols; it uses a different label source, but exact molecule-level disjointness from the primary panel was not established. It should therefore be interpreted as an exploratory label-source-shift transfer analysis only and does not address the methodological requirement for homogeneous endpoint-level analysis. Fourth, descriptor and kernel transfer analyses use different classifiers (Random Forest and SVM, respectively), so their AUCs are not a direct model-to-model comparison.

Fifth, the quantum kernel was evaluated on a classical statevector simulator with a 6-qubit IQPEmbedding; no hardware-level quantum advantage is claimed, and NISQ noise and sampling overhead would alter performance [27]. Sixth, the companion GNN study [28] provides architectural context, but cross-study comparisons remain

conditional on matching preprocessing, split definitions, and evaluation populations. Seventh, TNE loses 0.077 AUC under the ChEMBL label-source shift, while the Tartarus docking analysis tests information retention on the same molecular set rather than generalisation to novel compounds.

Eighth, the panel is scaffold-poor for its size. The 19 836 molecules resolve to 632 Bemis–Murcko scaffolds, or 324 once atom identity is stripped (a mean of 31 molecules per scaffold) and benzene alone is the core of 1 661 of them (8.4%). Random cross-validation therefore places analogues of the same scaffold on both sides of most fold boundaries, so the AUCs reported above describe interpolation within analogue series rather than generalisation to unseen chemotypes. Repeating the benchmark with folds grouped by scaffold, so that no scaffold appears in both training and test sets, costs all eight descriptors between 0.086 and 0.196 AUC (10 repeats of 5-fold grouped cross-validation, otherwise the protocol of Table 1). ECFP4 falls from 0.948 5 to 0.822 4 (2.5–97.5 percentile of repeat means 0.812 2–0.829 8), TFP from 0.869 1 to 0.725 3, and TNE from 0.720 5 to 0.634 2. The comparative conclusion is largely unaffected, ECFP4 remains the strongest descriptor and TNE the weakest under either split, but the ordering in between is not fully preserved: MACCS overtakes FCFP4, and TFP overtakes PHCO, which loses the most of any descriptor (0.196). Transferability across scaffolds is therefore not predicted by performance within them, and the absolute values reported under random splitting should not be read as estimates of performance on structurally novel chemotypes.

Ninth, the fusion comparison of Section 3.5 rests on fold-level tests that share training data ($df = 4$), so the stacked result is a failure to detect a difference rather than a demonstration that the representations are equivalent to ECFP4. It also speaks only to this endpoint: a descriptor that adds nothing to a fingerprint for predicted antiparasmodial activity may still add information for a different property.

4.6 Practical recommendations

Taken together, these results support ECFP4-RF as the primary screening baseline for the present panel (AUC 0.948); the ChEMBL analysis does not establish robustness to molecule-disjoint distribution shift. TFP is useful as an interpretable scaffold-preservation diagnostic, whereas TNE should be used only after transfer testing because its performance declined on the ChEMBL label-source shift. QKS is best regarded here as a controlled kernel comparator rather than a mechanistic assay. Future work should evaluate these representations on audited molecule-disjoint panels, experimental endpoints, and graph models using matched preprocessing and splits [29].

5 Conclusion

The central question was whether topological and quantum-inspired representations add value beyond ECFP4 for African antimalarial chemical space. We benchmarked persistent-homology TFP, Tucker-compressed TNE, and simulated QKS against classical baselines on computationally labelled data. TFP decomposes the scaffold paradox into conserved H_1 ring features and divergent H_0 connectivity. TNE achieves $6.1\times$

real-atom compression while retaining docking-score-predictive information, though its performance decreased under the ChEMBL label-source shift. The quantum kernel showed no performance advantage over a tuned RBF baseline under the tested protocols. These representations are complementary to classical fingerprints in the structural features they expose ring topology, compression behaviour, kernel geometry, but not in predictive information: stacking them onto ECFP4 changed AUC by -0.0002 (Section 3.5). They are diagnostic tools rather than replacements for classical screening baselines, with ECFP4 remaining the recommended primary descriptor. Persistent-homology decomposition provides an interpretable framework for evaluating generative-model outputs in natural-product drug discovery, while experimental validation on molecule-disjoint prospective panels remains necessary.

Supplementary information. Supplementary information accompanies this paper and includes detailed performance tables, bootstrap confidence intervals, additional topological analyses, ChEMBL transfer results, and quantum kernel benchmarks. The Supporting Information document is available as `Paper3_Quantum_Inspired_SM_V2609.pdf`.

Data Availability. All data, analysis scripts, and reproducibility records supporting this study are openly available at Zenodo: <https://doi.org/10.5281/zenodo.22768267> under the CC BY 4.0 license. The deposit contains: (1) TFP feature matrices for all 19849 library molecules; (2) TNE embeddings (192-element Tucker core descriptors, bond dimension 8); (3) quantum kernel matrices (6-qubit IQPEmbedding, PennyLane 0.45.1); (4) all benchmark CSV files; (5) the cross-paper H_1 /RRS analysis dataset; and (6) all analysis scripts with environment specifications for exact reproducibility.

List of Abbreviations. **TFP**: topological fingerprint; **TNE**: tensor network embedding; **QKS**: quantum kernel score; **ECFP4**: extended-connectivity fingerprint, radius 2; **AP**: atom-pair fingerprint; **BPF**: binding-pattern fingerprint; **FCFP4**: functional-class fingerprint, radius 2; **MACCS**: MACCS structural keys; **PHCO**: pharmacophore fingerprint; **$H_0/H_1/H_2$** : 0th/1st/2nd persistent homology dimensions; **AUC**: area under the receiver operating characteristic curve; **RBF**: radial basis function; **RRS**: resistance resilience score; **NISQ**: noisy intermediate-scale quantum; **UMAP**: uniform manifold approximation and projection; **RF**: random forest; **SVM**: support vector machine.

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Declarations

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Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

Consent for publication

Not applicable. No individual person’s data appear in this manuscript.

Author contributions

Myke Vital Sao Temgoua: conceptualization, methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapel Njafa: conceptualization, methodology, software, supervision, writing – review & editing. Serge Guy Nana Engo: methodology, supervision, writing – review & editing. Penabei Samafo: data curation, writing – review & editing. Fon Wilfred Mbacham: conceptualization, validation, writing – review & editing. All authors read and approved the final manuscript.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author.

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